Comparing Neural Models for AG News Topic Classification

Name: Yusuf Bulut
University ID: 3202960

Abstract

This project investigates whether explicit word order modeling improves neural topic classification on AG News enough to justify additional computational cost. I compare four models: a feed-forward neural network (FNN) that averages word embeddings, a simple RNN, an LSTM, and an LSTM with attention. The main task is 4-class AG News topic classification with the labels World, Sports, Business, and Sci/Tech. The main controlled comparison uses the same training, validation, and test subset sizes for all models. The FNN achieved the best final test accuracy, 0.8460, and the best macro F1, 0.8442, while also training fastest. The Attention LSTM was the strongest sequence model, with 0.8210 test accuracy and 0.8195 macro F1, but it did not outperform the simpler FNN. A sequence length analysis showed that increasing maximum length from 30 to 80 helped the FNN and Attention LSTM, but increasing from 80 to 150 did not improve test accuracy. Overall, these results suggest that AG News classification is strongly driven by topic-word signals, and that more complex sequence models are not automatically better for this task.

Research Question

On AG News topic classification, does modeling word order with RNN, LSTM, or Attention LSTM improve performance enough to justify extra training cost compared with a simple averaged-embedding FNN, and how does maximum sequence length affect this trade-off?

Dataset and Preprocessing

The dataset is AG News, a 4-class topic classification dataset with the labels World, Sports, Business, and Sci/Tech. AG News is commonly used as a text classification benchmark and is included in the benchmark collection used by Zhang et al. (2015). The commonly reported benchmark size is 120000 training examples and 7600 test examples, with 30000 training and 1900 test examples per class.

This project loads AG News through the Hugging Face `datasets` library (Lhoest et al., 2021). The raw dataset is not included in this repository or in the final ZIP. It is downloaded automatically on the first run through Hugging Face Datasets and cached locally. No private data or access token is required.

The Hugging Face metadata/source descriptions do not provide a clear standard licence and list the licence as unknown. Available descriptions state that the original AG corpus was provided for research and other non-commercial activity. Therefore, this project uses AG News only for an academic, non-commercial course project and does not redistribute the raw dataset.

The original training split is divided into training and validation data with a stratified split, and the original test split is used for testing. The final comparison uses subset limits of 10000 training examples, 2000 validation examples, and 2000 test examples. These limits keep the project feasible while still allowing a controlled model comparison.

Preprocessing is intentionally simple. Text is lowercased and tokenized with whitespace splitting. The vocabulary is built only from the training data. Two special tokens are used: `<pad>` for padding and `<unk>` for words outside the vocabulary. Texts are converted to token IDs, then padded or truncated to a fixed maximum length.

Methodology

The FNN is the baseline model. It embeds the input tokens, averages the non-padding token embeddings, and classifies the averaged representation. This model does not explicitly model word order, but it can learn strong topic-word signals.

The RNN processes tokens sequentially and uses the final non-padding hidden state for classification (Elman, 1990). The LSTM uses gates to model sequences more effectively than a simple RNN (Hochreiter and Schmidhuber, 1997). The Attention LSTM first encodes the sequence with an LSTM, then computes attention over the LSTM outputs before classification (Bahdanau et al., 2015).

The main evaluation uses accuracy and macro F1. Accuracy shows the overall proportion of correct predictions. Macro F1 is important because it averages class-level F1 scores equally across the four classes. I also record parameter count and training time to compare computational cost.

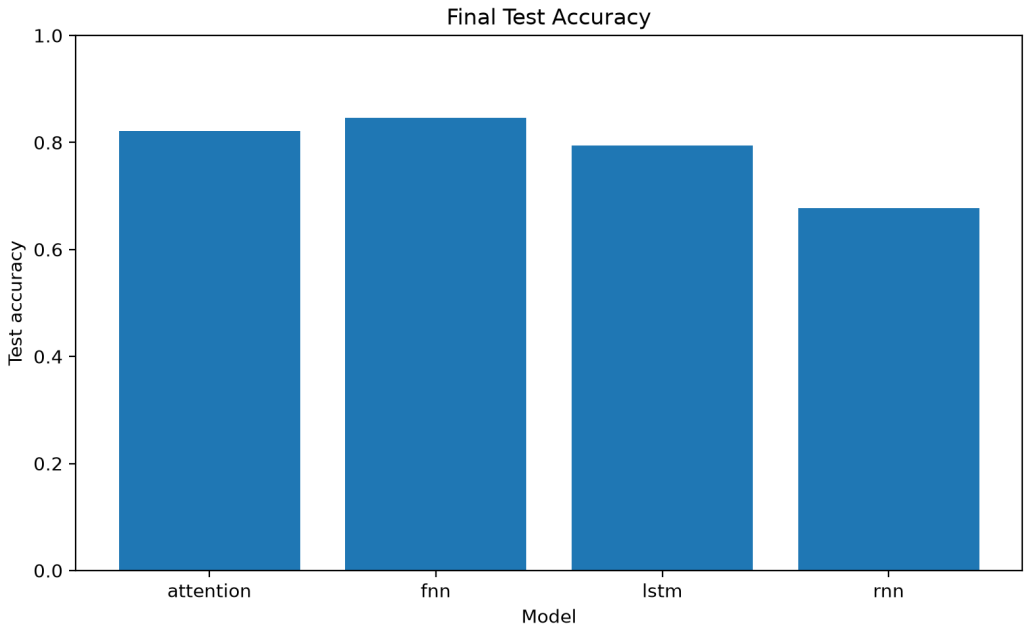
The main runs use seed 42, Python 3.14.6, PyTorch (Paszke et al., 2019), and the macOS MPS backend. The optimizer is Adam and the loss function is cross-entropy loss. The main experiment uses 5 epochs, batch size 32, maximum sequence length 80, and maximum vocabulary size 10000.

Main Results

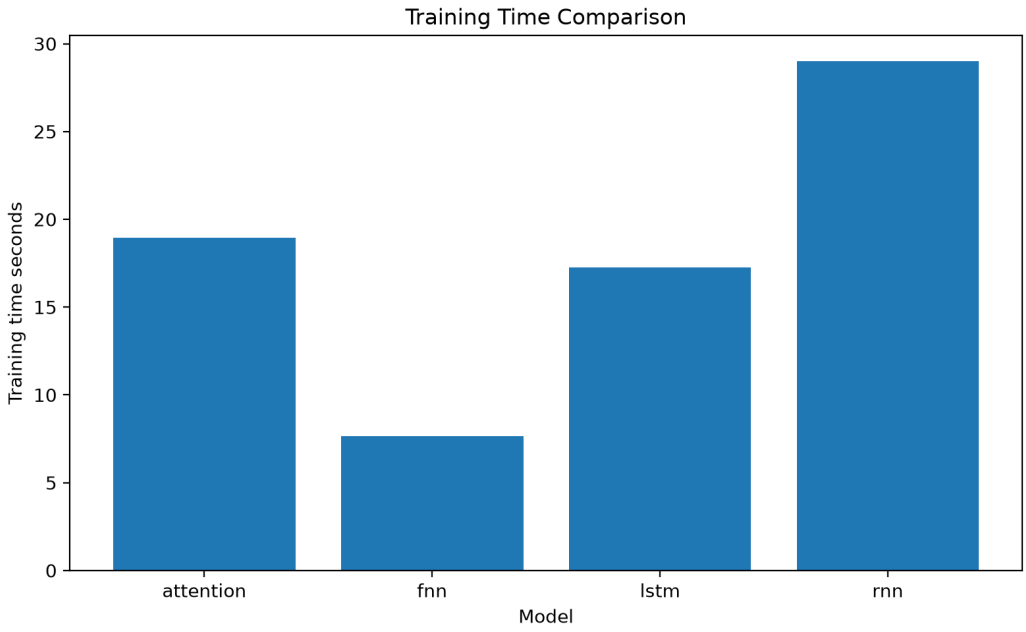
The detailed final results are saved in `results/final_metrics.csv` and summarized in `results/final_analysis_summary.md`.

Model	Validation accuracy	Test accuracy	Test macro F1	Parameters	Training time
FNN	0.8405	0.8460	0.8442	644420	7.64 s
RNN	0.7070	0.6775	0.6682	648580	29.02 s
LSTM	0.8100	0.7940	0.7928	673540	17.24 s
Attention LSTM	0.8125	0.8210	0.8195	673605	18.95 s

The FNN achieved the best validation accuracy, test accuracy, and macro F1. It was also the fastest model. The Attention LSTM was the best sequence model and clearly outperformed the plain LSTM, but it still did not beat the FNN. The simple RNN performed worst and also took the longest time in this run.



Final test accuracy by model.



Training time comparison by model.

Error Analysis

The per-class metrics and confusion matrices are saved under `results/analysis/`. They show that Sports is the easiest class across the models, while Business is the weakest class for all four models. The most systematic confusion is between Business and Sci/Tech.

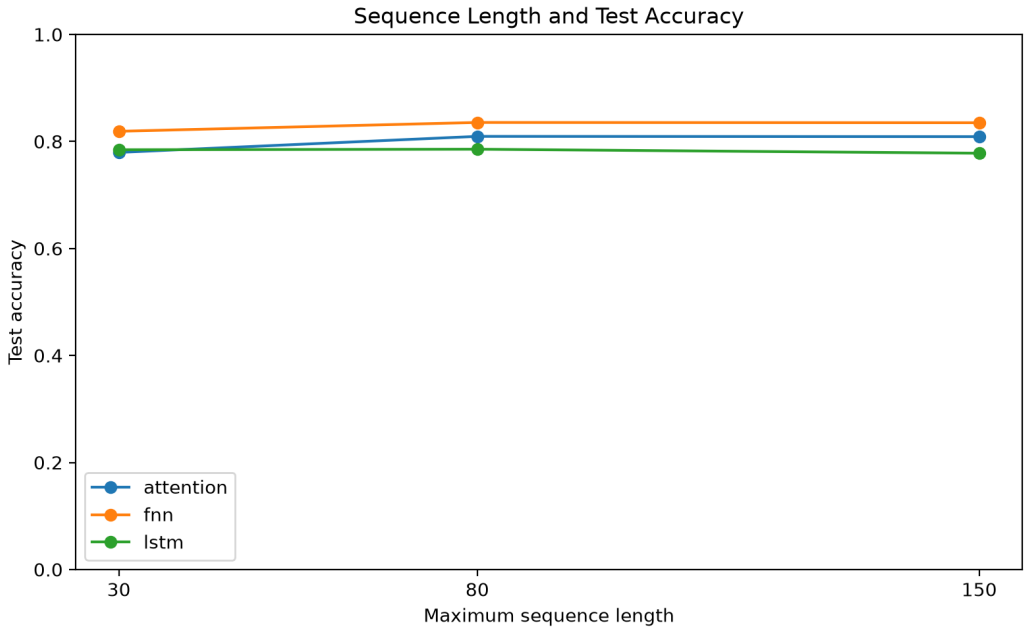
This confusion is understandable because many AG News articles contain mixed business and technology vocabulary. For example, technology-company stories can include company, market, product, acquisition, and finance language, while business articles can mention technical products or companies. World news can also overlap with Business when political events mention oil markets, trade, companies, or economic consequences. These errors suggest that the models often rely on topic words rather than deeper document interpretation. They also show that some AG News labels are naturally ambiguous when technology, business, politics, and science overlap.

Sequence Length Analysis

I also tested whether maximum sequence length affects performance. This analysis used FNN, LSTM, and Attention LSTM with maximum sequence lengths 30, 80, and 150. The results are saved in `results/sequence_length_summary.md`.

Model	Test accuracy at length 30	Test accuracy at length 80	Test accuracy at length 150
FNN	0.8190	0.8355	0.8350
LSTM	0.7845	0.7855	0.7780
Attention LSTM	0.7795	0.8095	0.8090

The FNN improved from length 30 to 80 and then plateaued. The Attention LSTM also improved from 30 to 80 and then plateaued. The LSTM did not clearly benefit from longer sequences. Increasing length to 150 added training cost without improving test accuracy.



Test accuracy for different maximum sequence lengths.

Discussion

The main finding is that the simplest model, the averaged-embedding FNN, performed best. This suggests that AG News topic classification is strongly driven by informative words such as team names, countries, companies, products, and topic-specific terms. In this setting, explicitly modeling word order was not necessary for the best result.

The sequence models still provide useful comparisons. The RNN was weak, which shows that a basic recurrent model is not automatically a good choice. The LSTM improved strongly over the RNN, and attention improved over the LSTM. However, the Attention LSTM required more parameters and more than twice the FNN training time while still achieving lower test accuracy.

The sequence length analysis supports the same interpretation. Some context is useful, because length 80 worked better than length 30 for the FNN and Attention LSTM. However, length 150 did not add a clear benefit, suggesting that the most useful topic information is often found in the shorter part of the article text.

Limitations

The experiments use subset limits of 10000 training examples, 2000 validation examples, and 2000 test examples, not the full AG News dataset. Only one random seed is reported. The tokenization is simple whitespace tokenization with lowercasing, so punctuation, morphology, and subword information are not modeled well. I did not include a pretrained Transformer baseline, and I did not do extensive hyperparameter tuning. Some AG News labels are ambiguous, especially between Business and Sci/Tech. Training time may vary depending on hardware and backend; these experiments were run on macOS with MPS.

Conclusion

This project compared FNN, RNN, LSTM, and Attention LSTM models for AG News topic classification. The FNN achieved the strongest final result with 0.8460 test accuracy and 0.8442 macro F1, while also training fastest. The Attention LSTM was the strongest sequence model, but it did not outperform the FNN. The main error pattern was confusion between Business and Sci/Tech, which reflects overlap between technology-company vocabulary and business vocabulary. Overall, the results show that simple averaged embeddings can be highly competitive for topic classification when class-specific keywords are informative.

Reproducibility

The code is in `src/`. Main final metrics are in `results/final_metrics.csv`. Per-class metrics and confusion matrices are in `results/analysis/`. The sequence length results are in `results/sequence_length_summary.md`. Raw-text prediction CSVs and full error-example Markdown files can be regenerated locally with `--save-analysis`, but they are excluded from the submitted package because they contain AG News text. The README contains setup and reproduction commands.

Declaration of AI Usage

I used ChatGPT and OpenAI Codex to help plan the project, generate and revise code, debug errors, design experiments, and draft parts of the report. I checked the generated code by running smoke tests, controlled experiments, and verification scripts, and I reviewed the reported results against the saved CSV files and summaries. I remain responsible for the submitted code, results, and discussion.

References

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